

Once
you stop
learning,
you start
dying.

—Albert Einstein

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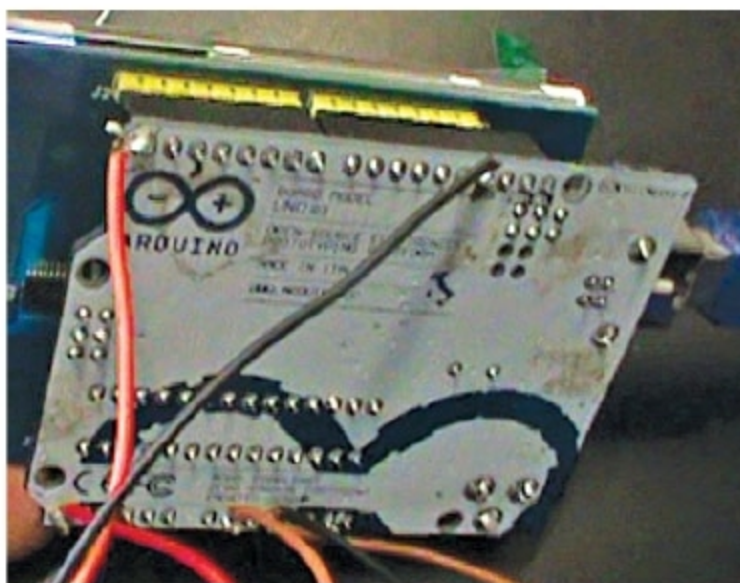


Fig. 3: Connections for GPS and TMP36



Fig. 4: Author's prototype

Construction

The project for the GPS analogue clock with temperature display requires following components:

1. 8.89cm TFT shield
2. Arduino Uno
3. U-Blox NEO-6M GPS receiver
4. LM35/TMP36 temperature sensor
5. 3.3V DC regulator/power supply

The connections are easy to make as shown in Fig. 2. The pin connections of the Arduino TFT shield are straightforward; you just need to mount the shield on top of the Arduino Uno board. The pin-to-pin connection details between TFT shield and Arduino Uno are given in the table.

Since top portion of the Arduino Uno is covered by TFT shield, the connections for the TMP36 and GPS receiver are taken from the bottom side of the Arduino Uno board (refer Fig. 3). In case you want to free some Arduino pins then you should see the `mcufriend_shield.h` file and re-write the connections to do so.

The shield is basically for 3.3V operation, but may work up to 5V. However, prolonged operation on 5V is not recommended as it gets overheated at that voltage. The 3.3V supply is derived from an LD1117V33 voltage regulator.

Software

Writing the Arduino code/sketch (`GPS_analog_clock.ino`) for the project is real fun! You can make TFT display in many different ways through coding. Creating a

thick border line, making the hour and minutes hands move smoothly were quite challenging as Adafruit_GFX library is not so much developed. High school trigonometry is all that you need to make it happen for you.

Open `GPS_analog_clock.ino` code from Arduino IDE and include the header files. Compile and upload the code to Arduino Uno board.

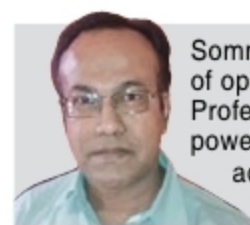
Another sketch (`GPS_analog_clock2.ino`) for changing colours of the minute hands every minute is also included.

Operation and testing

After uploading the code, solder the pins of TMP36 and GPS module to the Arduino board. Then mount the TFT shield on top of the Arduino board. After all the connections are done as per Fig. 2, connect it to 3.3V DC source. GPS module requires a few minutes to trace the satellites.

Most GPS receivers have an inbuilt patch antenna that can locate the low Earth orbit (LEO) GPS satellites very easily, even if your room's windows are closed. As soon as two such satellites are located, the time starts appearing on the analogue dial. At the same time, the date, time and temperature are shown digitally on right side of the TFT display.

Temperature sensor TMP36 works on 3.3V. An LM35 can also be used instead, but for that we need a 5V DC supply. **EFY**



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd

EFY Note

The source code of this project is available for free download at source.efymag.com